

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	Mohan Kumar J	Reg. No.:	192424060
Section / Batch:	AI 2 D8	Date:	07.08.2020

Instructions to Students

- Answer the following question. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1

SECTION A – CONCEPT MAPPING

1. Explain how algorithms compared using growth function. Extend map by including constraints & input characteristics. Justify importance of asymptotic analysis.

Code of Conduct

I Mohan Kumar (192424060) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Mohan Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

WHAT IS GROWTH FUNCTION?

A Function that describes how the time or space required by an algo grows as input size (n) increases.

ALGORITHM COST

Cost is measured as function of input size n .
Common measures:
Time complexity $T(n)$
Space complexity $S(n)$

GROWTH FUNCTIONS

Constant: $O(1)$
Logarithmic: $O(\log n)$
Linear: $O(n)$
Linearithmic: $O(n \log n)$
Quadratic: $O(n^2)$
Factorial: $O(n!)$
Exponential: $O(2^n)$

HOW ALGORITHM COMPARED

We compare $T_1(n)$ & $T_2(n)$ using Growth functions

1. If $T_1(n) \in O(T_2(n))$
 $T_2(n)$ grows no faster than $T_1(n)$
2. If $T_1(n) \in \Omega(T_2(n))$
 $T_1(n)$ grows at least as fast as $T_2(n)$
3. If $T_1(n) \in \Theta(T_2(n))$
Both grows at same rate

INPUT CHARACTERISTIC

Input size (n): No. of elements
Input Distribution: Sorted, reverse sorted, random, etc..
Data Types & Range: Integer, real, characters, etc..
Structure: Array, linked list, tree, graph, etc..
Density: Sparse or Dense

CONSTRAINTS

Time constraint
Space constraint
Real-time constraint
Hardware constraint
Power constraints

ASYMPTOTIC ANALYSIS

Focus on growth rate
 $n \rightarrow \infty$
Ignore constant & lower order terms
Provides theoretical upper bound
Predicts performance/scalability

WHY ASYMPTOTIC ANALYSIS IS IMPORTANT?

Simplifies complex mathematical expressions
Helps compare algo independent of hardware
Predicts performance for large input sizes
Essential in theoretical Computer Science & Practical Application

HOW THIS MAPPING LEADS TO OPTIMAL ALGO SELECTION
Understand input characteristics & constraints of Problem
Analyze candidate algorithms using asymptotic growth function
Compare their growth rates using O , Ω , Θ notations
Choose algorithm with best (lowest) growth rate satisfies all constraints.